ECE 341

Homework 1

The problem numbers listed below are from the Lathi textbook, 3rd edition.

Prob. Number	Course Outcome	Points
1.1-1	1-b	12
1.1-4	1-b	12
1.2-1	1-b	16
1.4-2	1-b	14
1.5-4	1-b	12
1.7-2	1-b	18
1.7-15	1-b	16
	TOTAL:	100

1-b. Evaluate and classify signals to identify unstable systems.

1.1.1 a) $x_a(t)$'s energy is given by

$$E_a = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^2 1dt + \int_2^3 1dt = 2 + 1 = 3.$$

b) $x_b(t) = -x_a(t)$, and its energy is given by

$$E_b = \int_0^2 1dt + \int_2^3 1dt = 3.$$

c) $x_c(t) = x_a(t-3)$, and its energy is given by

$$E_c = \int_3^5 1dt + \int_5^6 1dt = 3.$$

d) $x_d(t) = 2x_a(t)$, and its energy is given by

$$E_d = 4(\int_0^2 1dt + \int_2^3 1dt) = 4(3) = 12.$$

Since a signals energy is based off the absolute value of the signal, $|x(t)|$, sign changes have no effect on the energy, since $|-x(t)| = x(t)$. Additionally, since energy is the sum of all values of $t \in (-\infty, \infty)$, time-shifting has no effect on the energy. Finally, since energy is based off the square $|x(t)|^2$ of the given signal, doubling the signal will increase its energy by a factor of $2^2 = 4$, and multiplying the signal by k will increase its energy by a factor of k^2. In other words,

$$E_{kx(t)} = k^2 E_{x(t)}.$$

1.1.4 The signal shown in Fig. P1.1-4 is periodic with a period of $T = 4$, and thus has an energy of 0; therefore, power is a more meaningful measure of the signal's energy. Its power is given by

$$P_x = \frac{1}{4}\int_{-2}^{2} |x(t)|^2 dt = \frac{1}{4}\int_{-2}^{2} t^6 dt \approx 9.14.$$

The RMS value is given by the square root of the power,

$$P_{rms} = \sqrt{9.14} \approx 3.02.$$

a) The power of $x(t) = -x(t) = -t^3$ is found as

$$P_{-x} = \frac{1}{4}\int_{-2}^{2} |-t^3|^2 dt \approx 9.14,$$

with an RMS value of $\sqrt{9.14} \approx 3.02$.

b) The power of $x(t) = 2x(t) = 2t^3$ is found as

$$P_{2x} = \frac{1}{4}\int_{-2}^{2} |2t^3|^2 dt = \frac{1}{4}\int_{-2}^{2} 4t^6 dt \approx 36.57,$$

with an RMS value of $\sqrt{36.57} \approx 6.05$.

c) The power of $x(t) = cx(t) = ct^3$ is found as

$$P_{cx} = \frac{1}{4}\int_{-2}^{2} |ct^3|^2 dt = \frac{1}{4}\int_{-2}^{2} c^2t^6 dt = c^2(9.14),$$

with an RMS value of $\sqrt{c^2(9.14)} = c(3.02)$.

1.2.1 The given signal, along with its permutations, are labeled on the y-axis and shown below (graphs were created in MATLAB from scratch):

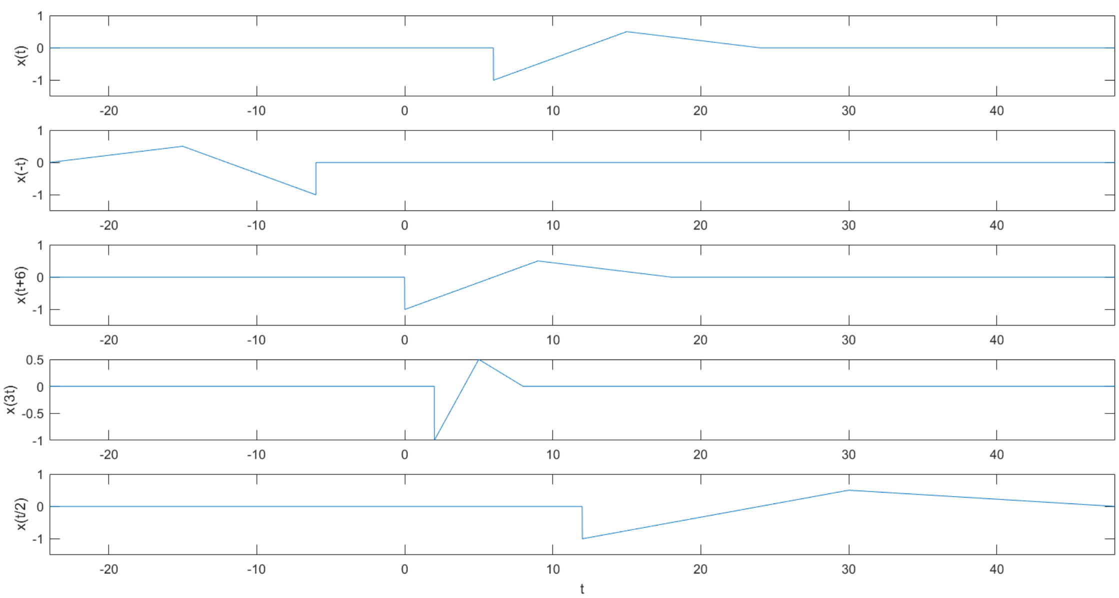

Figure 1: $x(t)$ and its permutations

1.4.2 a) $x_1(t) = (4t+4)(u(t+1) - u(t)) + (-2t+4)(u(t) - u(t-2))$.

b) $x_2(t) = t^2(u(t) - u(t-2)) + (2t-8)(u(t-2) - u(t-4))$.

1.5.4 a) The product of an even and an odd function equals an odd function, $x_e(t)x_o(t) = x_o(t)$, and the integral of an odd function over any symmetric bound $(-a, a)$is equal to 0, as the negative and positive components cancel out. Mathematically,

$$\int_{-\frac{a}{2}}^{0} x_o(t)dt + \int_{0}^{\frac{a}{2}} x_o(t)dt = 0.$$

Therefore, the integral of the product $x_o(t)x_e(t) = x_o(t)$ is always equal to 0 on the interval $(-\infty, \infty)$.

b) Since the integrals of odd functions on symmetric bounds, as mentioned above, always evaulate to 0, and all functions $x(t)$ are the sum of an even and odd component $x_o(t) + x_e(t)$, we can express the integral of any function $x(t)$ on the bound $(-\infty, \infty)$ as

$$\int_{-\infty}^{\infty} x(t)dt = \int_{-\infty}^{\infty} x_o(t) + \int_{-\infty}^{\infty} x_e(t)dt = 0 + \int_{-\infty}^{\infty} x_e(t)dt.$$

1.7.2 For a time-invariant system $y(t) = x(t)$, supposing $x_1(t) = x(t)$ and $y_1(t) = y(t)$, time shifting the input such that $x_2(t) = x_1(t - t_0)$ should result in an equivalent shift of $y_2(t) = y_1(t - t_0)$, so that $y_1(t) = x_1(t)$ and $y_2(t) = x_2(t)$. If this is not the case, the system is considered time-variant.

a) Given $x_1(t) = x(t-2)$, $x_2(t) = x_1(t-t_0) = x(t-t_0-2)$. Then, shifting the output by the same time delay, $y_2(t) = y_1(t - t_0) = x_1(t - t_0 - 2)$.
$y_2(t) = x(t-t_0-2) = x_2(t)$, proving that the system is time invariant.

b) Given $x_1(t) = x(-t)$, $x_2(t) = x_1(t - t_0) = x(-t - t_0)$. Then, shifting the output by the same time delay, $y_2(t) = y_1(t-t_0) = x(-(t-t_0)) = x(-t + t_0)$. .
$y_2(t) = x(-t + t_0) \neq x(-t - t_0) = x_2(t)$, proving that the system is time variant.

c) Given $x_1(t) = x(at)$, $x_2(t) = x_1(t - t_0) = x(at - t_0)$. Then, shifting the output by the same time delay, $y_2(t) = y_1(t - t_0) = x(a(t - t_0)) = x(at - at_0)$.
$y_2(t) = x(at - at_0) \neq x(at - t_0) = x_2(t)$, proving that the system is time variant.

d) Given $x_1(t) = tx(t - 2)$, $x_2 = x_1(t - t_0) = tx(t - t_0 - 2)$. Then, shifting the output by the same time delay, $y_2(t) = y_1(t - t_0) = (t - t_0)x(t - t_0 - 2)$.
$y_2(t) = (t - t_0)x(t - t_0 - 2) \neq tx(t - t_0 - 2) = x_2(t)$, proving that the system is time variant.

e) Given $y(t) = \int_{-5}^{5} x(\tau)d\tau$, a simple counterexample may be used to show that the system is time variant.
Assuming that $x_1(t) = \delta(t)$, then

$$y_1(t) = \int_{-5}^{5} \delta(\tau)d\tau = 1.$$

Assuming that $x_2(t) = \delta(t-10)$, then

$$y_2(t) = \int_{-5}^{5} \delta(\tau) d\tau = 0.$$

Since the output of $y_1(t) = 1$ is a constant (does not depend on t), in a time-invariant system, we would expect $y_2(t) = y(t-10) = 1$. However, $y_2(t) = 1 \neq 0 = x_2(t)$, and so the system is time variant.

f) Given $x_1 = (\frac{dx(t)}{dt})^2$, $x_2(t) = x_1(t-t_0) = (\frac{dx(t-t_0)}{dt})^2$. Then, shifting the output by the same amount, $y_2(t) = y_1(t-t_0) = (\frac{dx(t-t_0)}{d(t-t_0)})^2$. Note that dt and $d(t-t_0)$ have the same value, as they are infintesimal quantites, and thus we can say $dt = d(t-t_0)$. Therefore, $y_2(t) = \frac{dx(t-t_0)}{dt} = x_2(t)$, proving the system is time invariant.

1.7.15 a) For $y(t) = x(t-2)$, the system is causal, since $t > t-2$ for all values of t, and therefore $y(t)$ depends on future values of t regardless of the input.

b) For $y(t) = x(-t)$, the system is noncausal, since, for negative values of $t = -t$, $-t < -(-t) = t$, making $y(t)$ dependent on a future value of t for $t < 0$.

c) For $y(t) = x(at)$, $a > 1$, the system is noncausal, since $t \leq at$ for all values of t and $a > 1$, making $y(t)$ dependent on future values of t for $a \neq 1$.

d) For $y(t) = x(at)$, $a < t$, the system is noncausal, since $t < at$ for all values of $t < 0$ and $a < 1$, making the system dependent on future values of t for $t < 0$.